

CoLA: Compute-Efficient Pre-Training of LLMs via Low-Rank Activation

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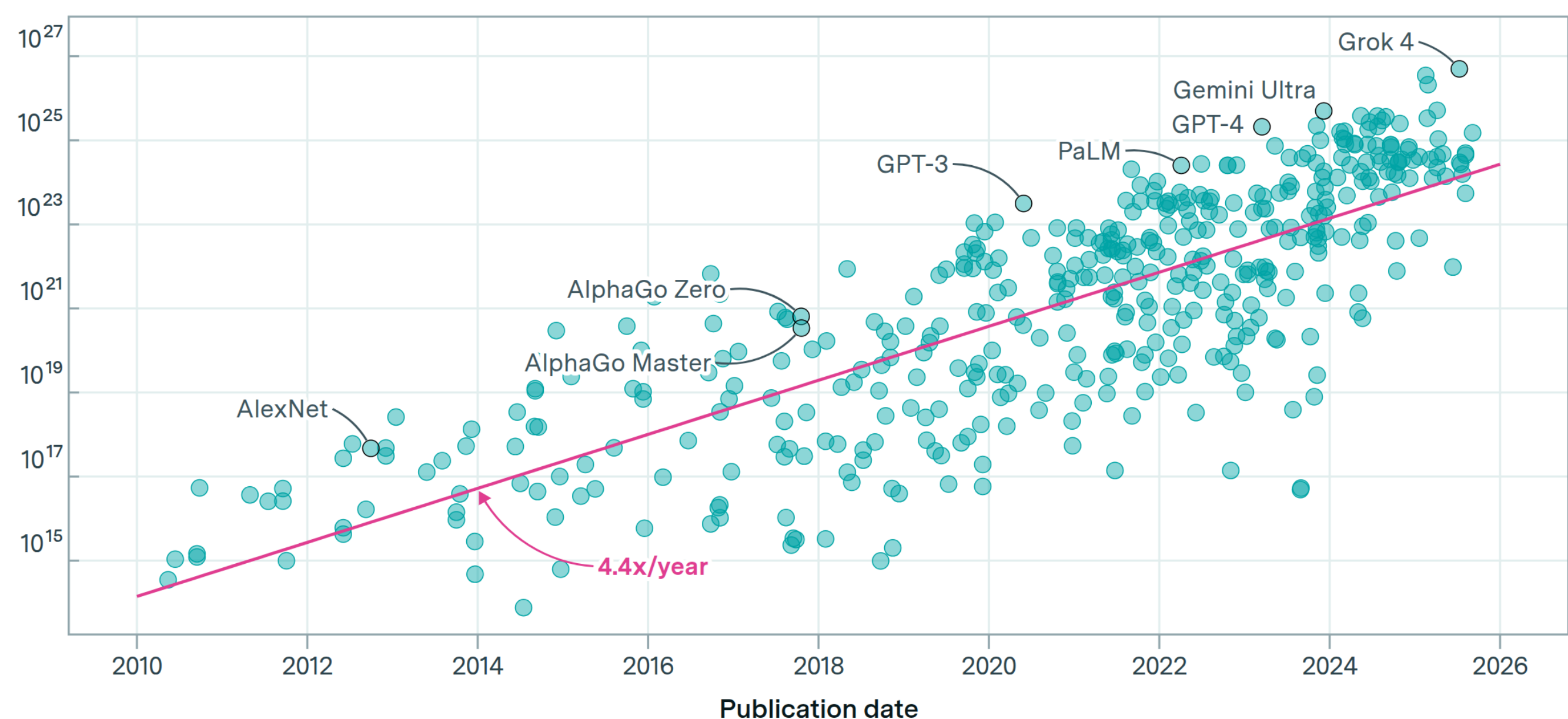
Background – LLM Pre-Training

Training compute of notable models

EPOCH AI

Training compute (FLOP)

443 models

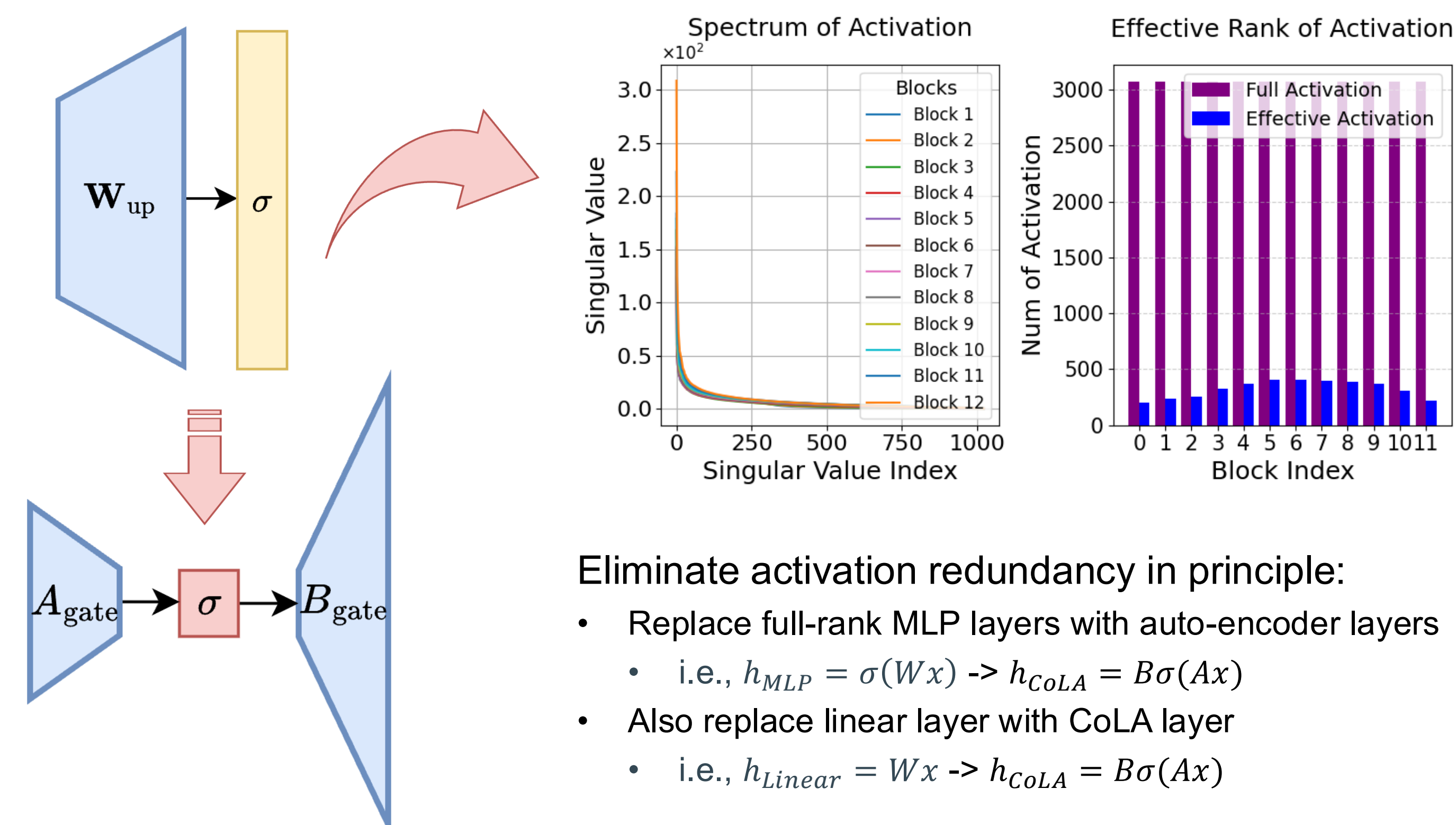


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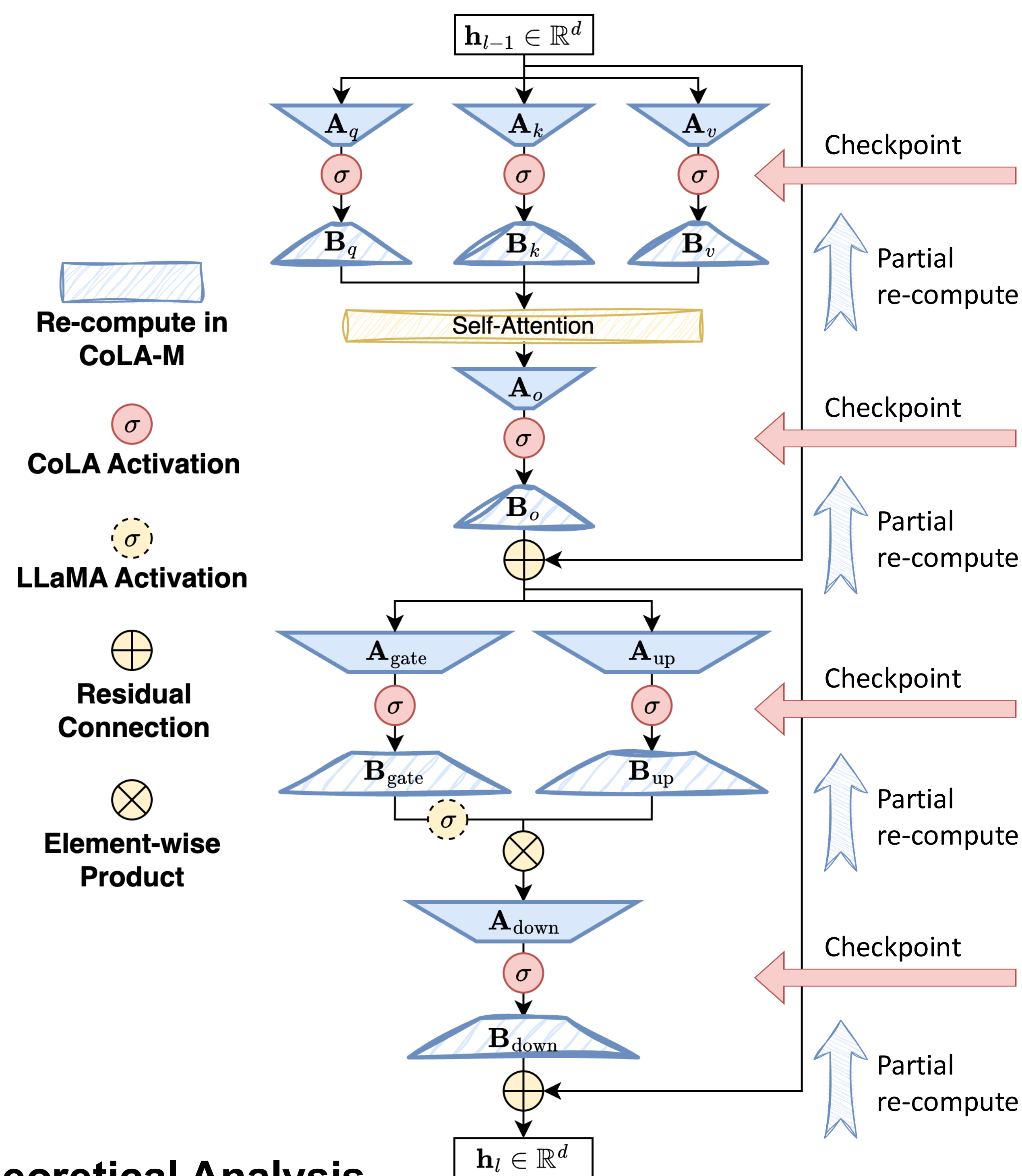
Observation – Low-Rank Activation

Activation function exhibits low *effective* rank



CoLA for Efficient LLM Pre-Training

We propose CoLA for **Efficient Training via Low-Rank Activation**.



Theoretical Analysis

Result 1 (Proposition 3.1): The approximation provided by CoLA layer is guaranteed to be no worse than the standard low-rank layer.

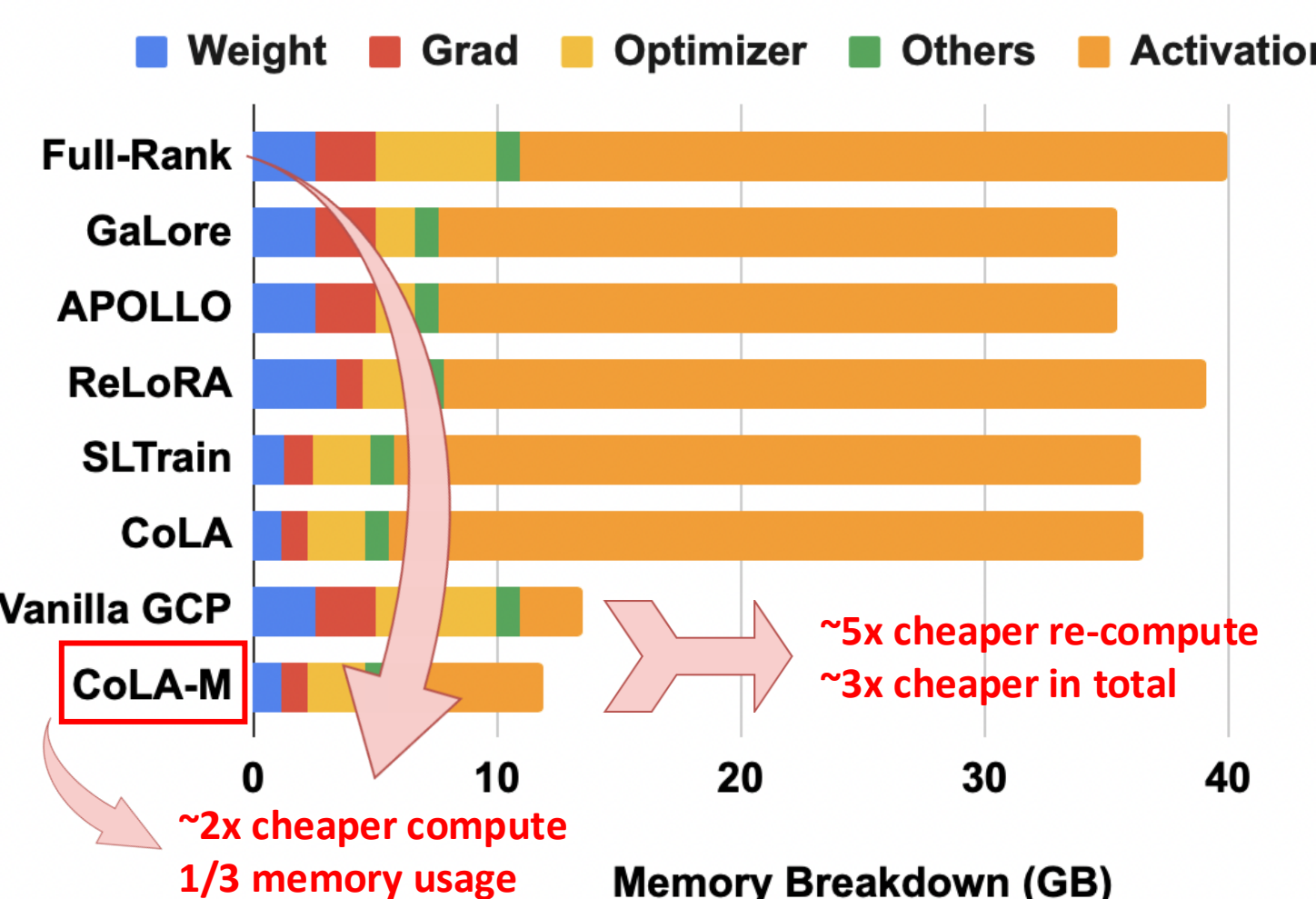
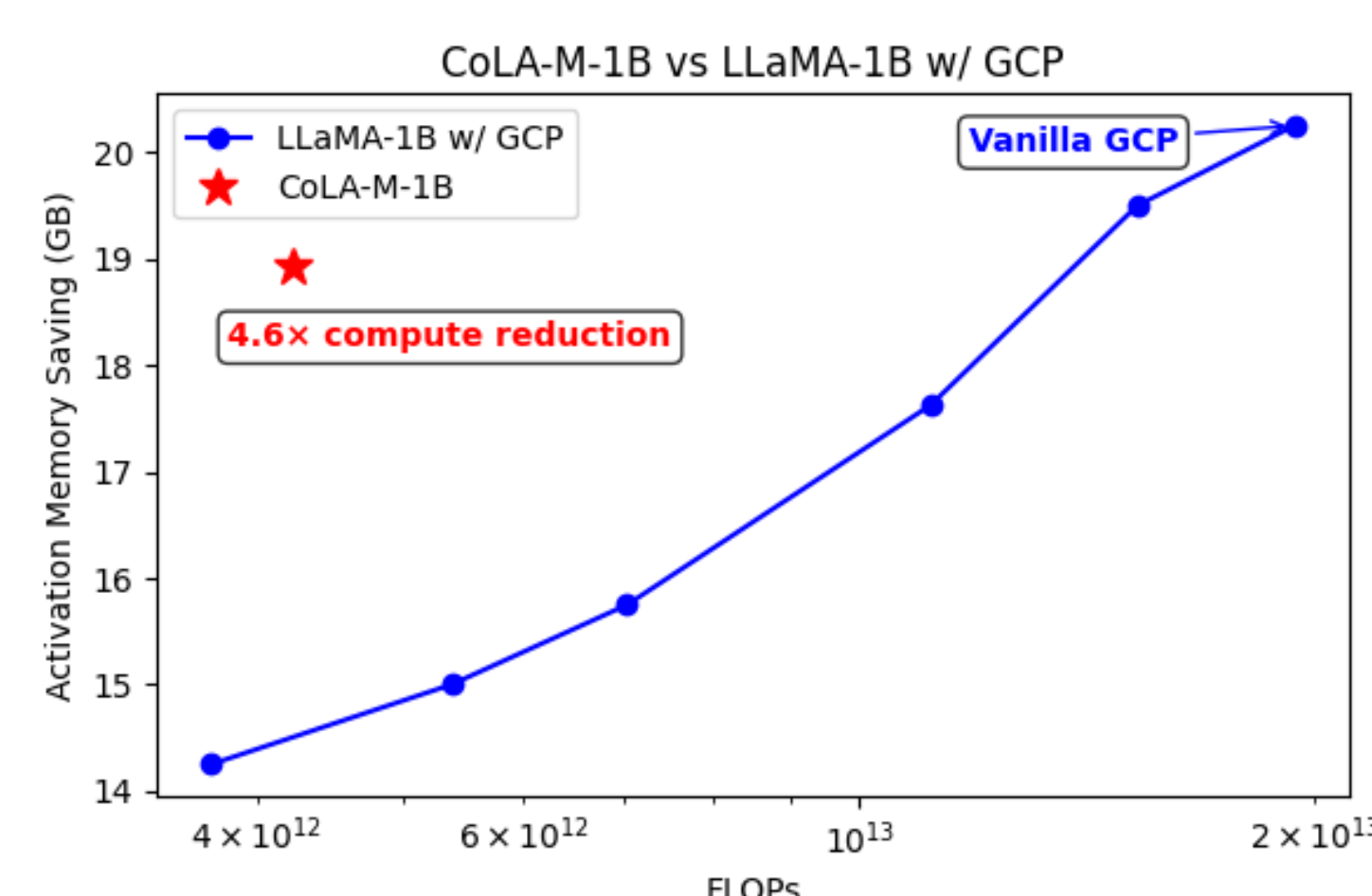
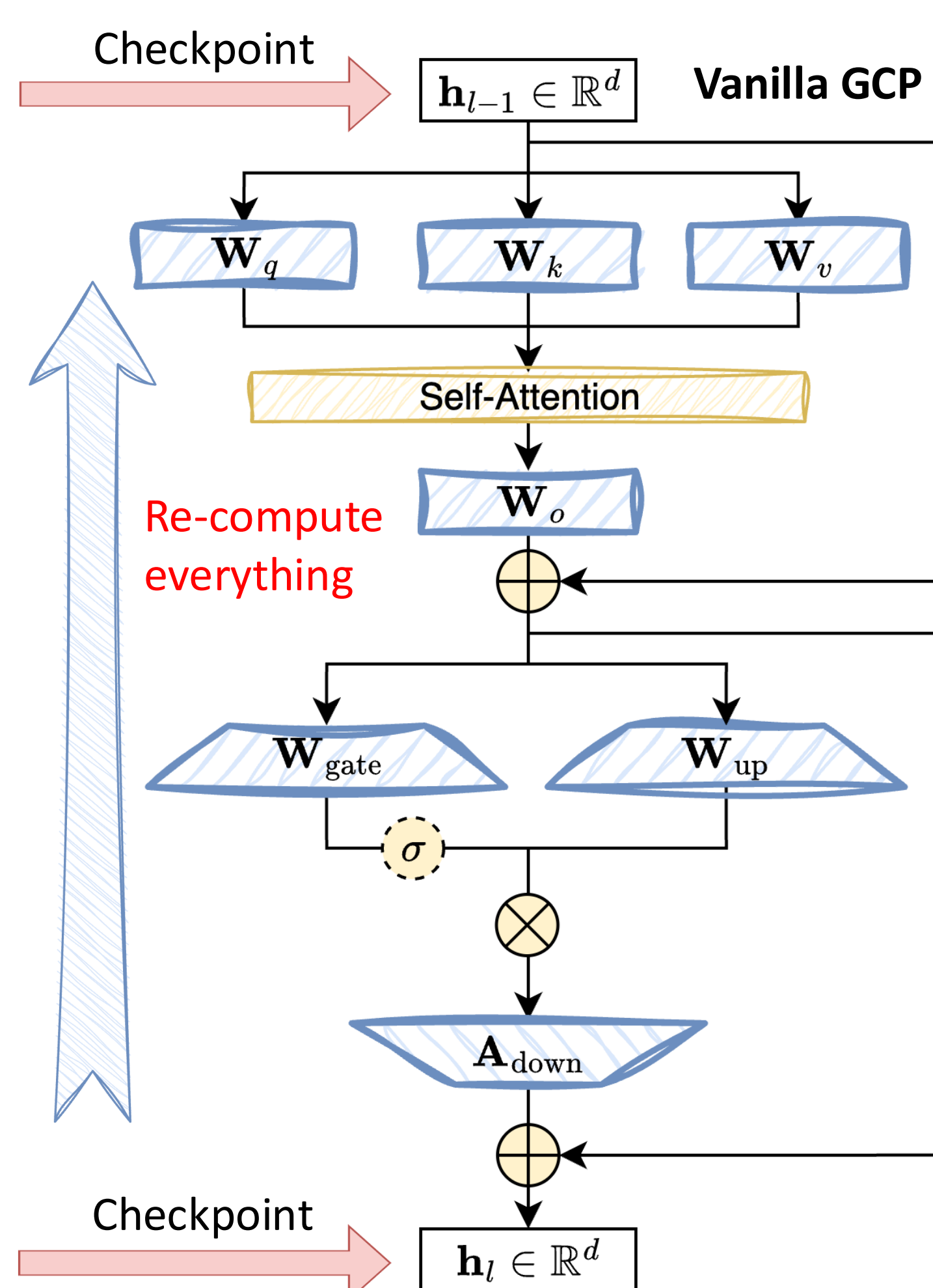
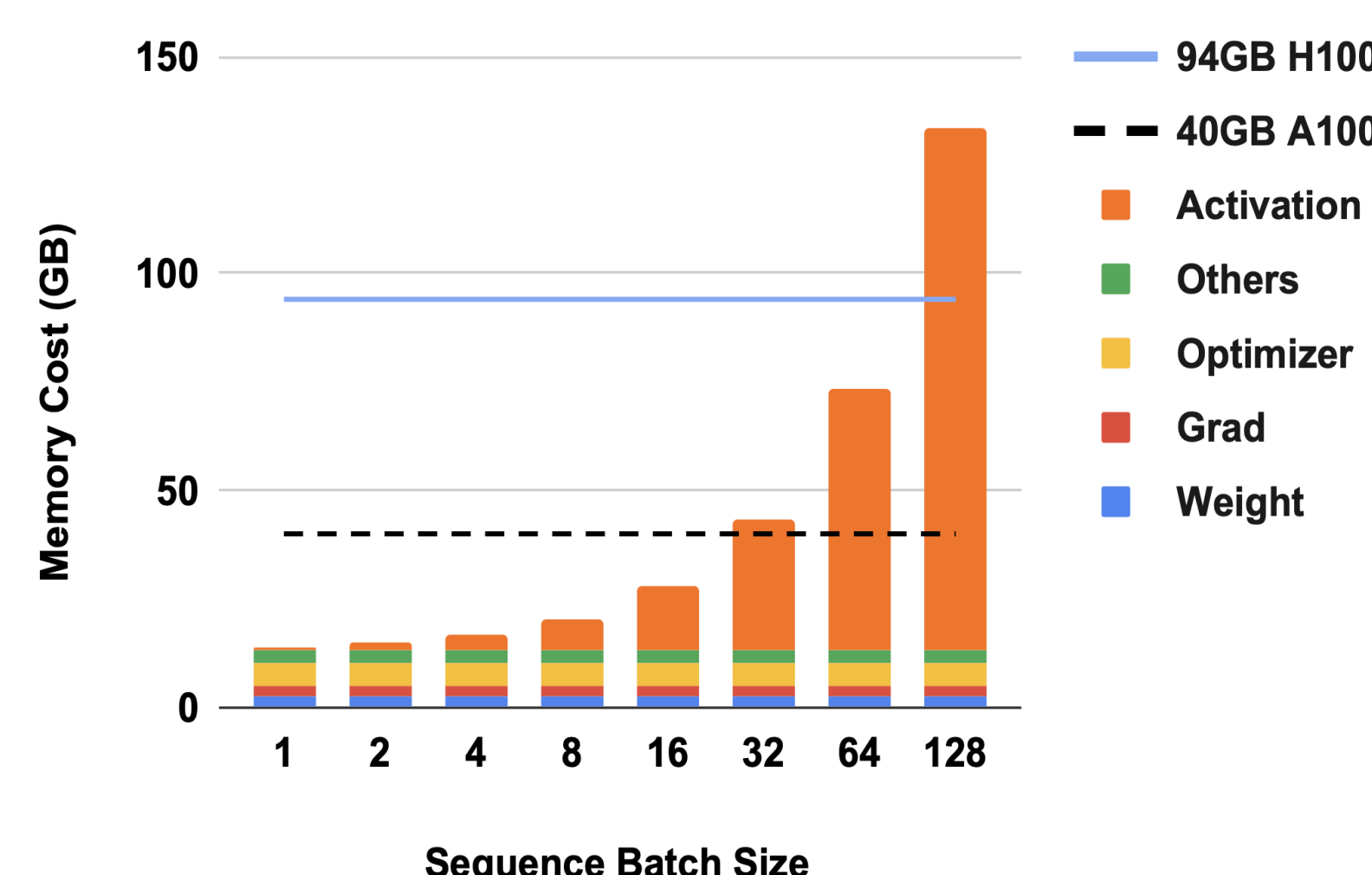
Result 2 (Theorem 3.3): Under suitable data-dependent conditions, CoLA layer strictly outperforms the standard low-rank layer.

Result 3 (Theorem 3.4): The CoLA layer provably achieves an error bound that adapts to the effective rank of the output, giving significantly tighter guarantees when the spectrum is concentrated

CoLA-M: Pushing Memory Efficiency

Activation memory can be substantial or dominating

- Gradient Checkpointing is often used in practice
 - Introduce significant re-computation cost
- CoLA-M achieves efficient gradient checkpointing
 - Checkpoint at low-rank dimensions
 - Reduce re-computation to half a CoLA forward
 - ~5x cheaper than full-rank GCP



Experiments – Pre-Training within Compute-Optimal

CoLA performs on par with full-rank models with being 2x smaller

	60M			130M			350M			1B		
r/d	128 / 512			256 / 768			256 / 1024			512 / 2048		
Tokens	1.1B			2.2B			6.4B			13.1B		
	PPL	Param (M)	Mem (GB)	PPL	Param (M)	Mem (GB)	PPL	Param (M)	Mem (GB)	PPL	Param (M)	Mem (GB)
Full-rank	34.06	58	0.43	24.36	134	1.00	18.80	368	2.74	15.56	1339	9.98
ReLoRA	37.04	58	0.37	29.37	134	0.86	29.08	368	1.94	18.33	1339	6.79
GaLore	34.88	58	0.36	25.36	134	0.79	18.95	368	1.90	15.64	1339	6.60
SLTrain	34.15	44	0.32	26.04	97	0.72	19.42	194	1.45	16.14	646	4.81
CoLA	34.04	43	0.32	24.48	94	0.70	19.40	185	1.38	15.52	609	4.54

For LLaMA 7B:

- CoLA-M with BF16 Adam consumes only ~1/3 memory than full-rank with 8-bit Adam
- CoLA achieves lowest perplexity

	Mem (GB)	10k	40k	80k	120k	150k
8-bit Adam	72.59	N/A	18.09	15.47	14.83	14.61
8-bit GaLore	65.16	26.87	17.94	15.39	14.95	14.65
SLTrain	60.91	27.59	N/A			
CoLA-M	26.82	22.76	16.21	13.82	13.09	12.73

~1/3 memory Lower perplexity

Experiments – Pre-Training beyond Compute-Optimal

LLaMA-350M with 51B tokens: CoLA outperforms LLaMA by **13.96** vs 14.47

BERT-350M with 85B tokens: CoLA outperforms BERT on GLUE by **83.5** vs 82.7

	Pre-Training Loss	QQP	SST-2	MRPC	COLA	QNLI	MNLI	RTE	STS-B	GLUE Avg
BERT _{Large}	1.263	91.1	92.1	90.7	53.1	91.6	84.3	69.9	88.9	82.7
CoLA	1.257	91.2	92.3	90.6	54.1	91.7	84.3	74.2	89.7	83.5

Experiments – System Performance

CoLA off-the-shelf can speed up training and inference

For training LLaMA 1B:

- CoLA and CoLA-M are the only two that are faster than full-rank baseline
- Meanwhile, CoLA-M reduces memory and compute by ~3x and ~2x
- We observe similar benefit for inference

